PROJECT REPORT

Urban Traffic Accidents Analysis

Tableau Dashboard — Comprehensive Safety Intelligence Report

Prepared: April 2026 | Tool: Tableau | Domain: Road Safety Analytics

1. Executive Summary

This report presents a comprehensive analysis of urban traffic accidents across India, built using Tableau. The dashboard provides multi-dimensional visibility into accident patterns, fatality rates, geographic hotspots, and causal factors. Decision-makers can leverage these insights to prioritise road safety interventions, allocate resources, and formulate evidence-based traffic policies.

2. Project Overview

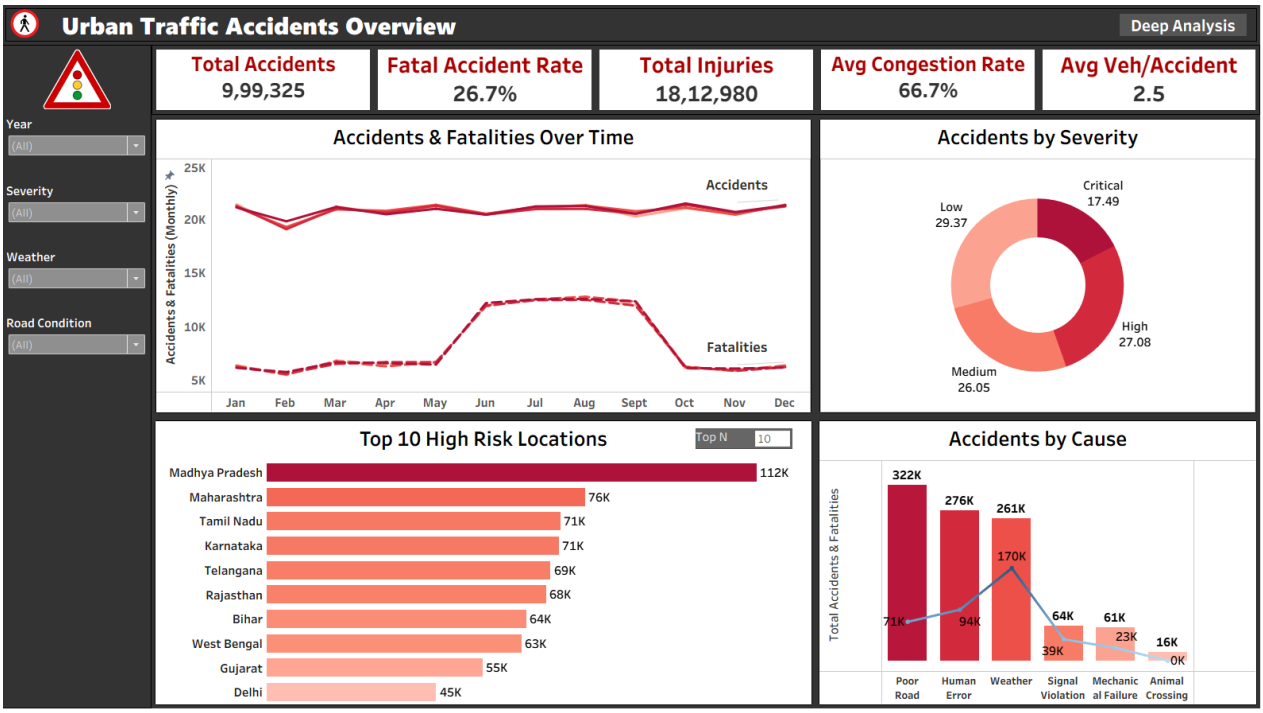
Project Title	Urban Traffic Accidents Analysis
Tool / Platform	Tableau (.twbx)
Dashboard Pages	2 — Overview & Cause/Risk Analysis
Data Scope	Pan-India, All Years, All Severities
Primary Audience	Traffic authorities, Policy makers, Urban planners

3. Key Performance Indicators

The following headline metrics are surfaced on the Overview dashboard:

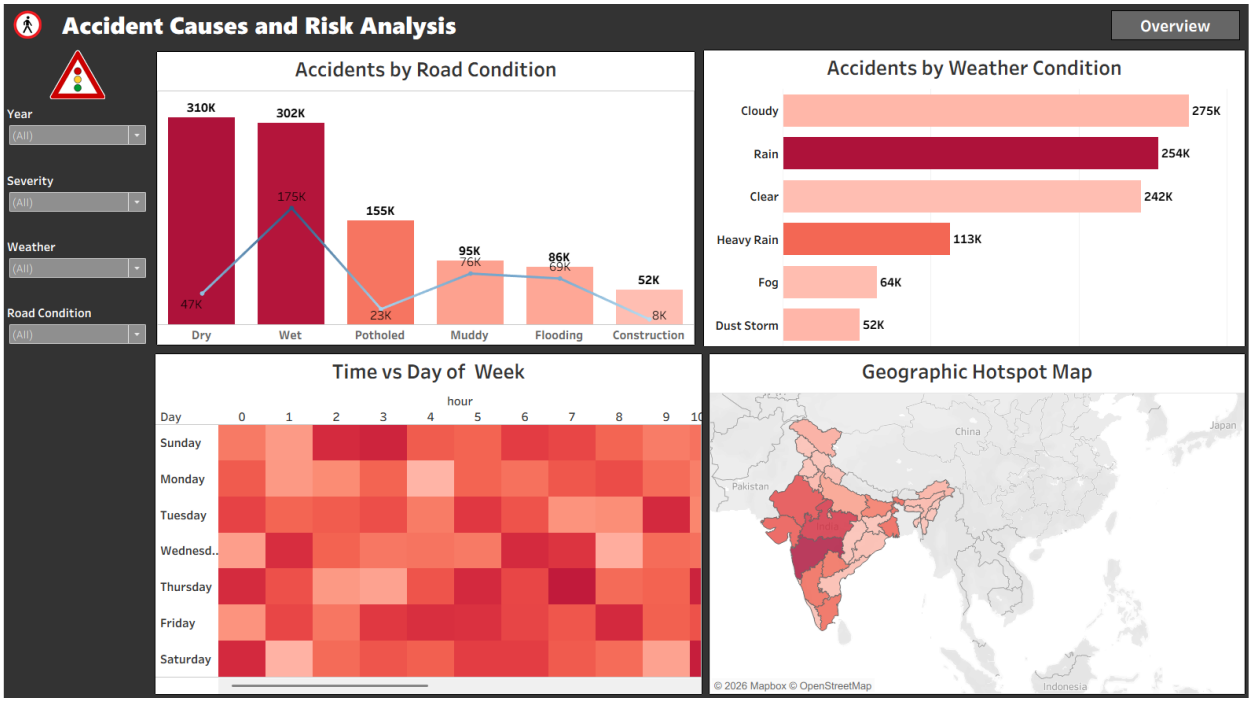
Total Accidents	9,99,325
Fatal Accident Rate	26.7%
Total Injuries	18,12,980
Avg Congestion Rate	66.7%
Avg Vehicles per Accident	2.5

4. Dashboard Structure & Visualisations



4.1 Overview Dashboard

- Accidents & Fatalities Over Time (dual-line chart): Monthly trend from January to December. Accidents remain relatively stable (~20K/month); fatalities spike sharply June–September, peaking near 12K, before falling in Q4.
- Accidents by Severity (donut chart): Low 29.37% | Medium 26.05% | High 27.08% | Critical 17.49%. Nearly half of all accidents fall in high or critical categories.
- Top 10 High-Risk Locations (bar chart): Madhya Pradesh leads with 112K accidents, followed by Maharashtra (76K), Tamil Nadu and Karnataka (71K each). The top 3 states account for over 25% of all incidents.
- Accidents by Cause (combo chart): Poor Road conditions are the #1 cause (322K), followed by Human Error (276K) and Weather (261K). Signal Violation contributes 64K, Mechanical Failure 61K, and Animal Crossing 16K.



4.2 Accident Causes & Risk Analysis Dashboard

- Accidents by Road Condition (bar+line combo): Dry roads account for 310K accidents — highest — closely followed by Wet roads (302K), indicating that even non-adverse conditions pose significant risk. Potholed (155K), Muddy (95K), Flooding (86K).
- Accidents by Weather Condition (horizontal bar): Cloudy weather sees the most accidents (275K), then Rain (254K) and Clear (242K). Heavy Rain (113K), Fog (64K), and Dust Storm (52K) follow.
- Time vs Day of Week Heatmap: Identifies peak risk windows — Thursday and Tuesday afternoons (hours 5–8) appear particularly dense. Saturdays show elevated risk in early morning hours.
- Geographic Hotspot Map: State-level chloropleth over India; Central and South Indian states (Madhya Pradesh, Maharashtra, Tamil Nadu) appear darkest, indicating highest accident density.

5. Filters & Interactive Controls

Both dashboards share a consistent filter panel on the left sidebar:

- Year — filter by individual year or all years
- Severity — Low / Medium / High / Critical
- Weather — all weather condition types
- Road Condition — Dry, Wet, Potholed, Muddy, Flooding, Construction
- Top N — dynamically controls how many top risk locations are shown (default: 10)
- Deep Analysis / Overview navigation buttons — seamless page switching

6. Key Insights & Findings

- Monsoon Effect: Fatalities spike dramatically between June and September, aligning with the Indian monsoon season, suggesting weather-related driving risks are severely underestimated.
- Dry Road Paradox: Despite dry roads being considered safer, they top accident counts (310K), possibly due to higher speeds and complacency on perceived safe surfaces.
- Central India Risk Concentration: Madhya Pradesh alone accounts for 11% of all accidents, making it the single highest-priority intervention zone.
- Human Error vs Infrastructure: Human Error (276K) and Poor Road conditions (322K) together account for nearly 60% of all accidents — both are addressable through targeted policy.
- Congestion as a Proxy: An average congestion rate of 66.7% indicates severely stressed urban road infrastructure across the country.

7. Recommendations

- Deploy seasonal safety campaigns and enforce reduced speed limits in monsoon months (June–September).
- Prioritise road resurfacing and pothole repair programs in Madhya Pradesh, Maharashtra, and Tamil Nadu.
- Enhance driver education on high-speed dry-road risks and signal compliance.

- Install smart traffic monitoring systems at hotspot intersections identified in the heatmap.
- Introduce mandatory vehicle maintenance checks to reduce mechanical failure-related accidents.

8. Technical Notes

File Format	Tableau Packaged Workbook (.twbx)
Visualisation Types	Line chart, Donut, Bar, Combo, Heatmap, Choropleth map
Map Provider	Mapbox / OpenStreetMap
Color Theme	Dark background with red-gradient severity encoding
Calculated Fields	Fatal Accident Rate, Avg Congestion Rate, Avg Veh/Accident

9. Conclusion

The Urban Traffic Accidents Analysis dashboard delivers actionable intelligence for road safety stakeholders across India. By combining temporal, geographic, and causal dimensions, it enables data-driven prioritisation of interventions. Continued data collection and quarterly dashboard refreshes will amplify its impact over time.